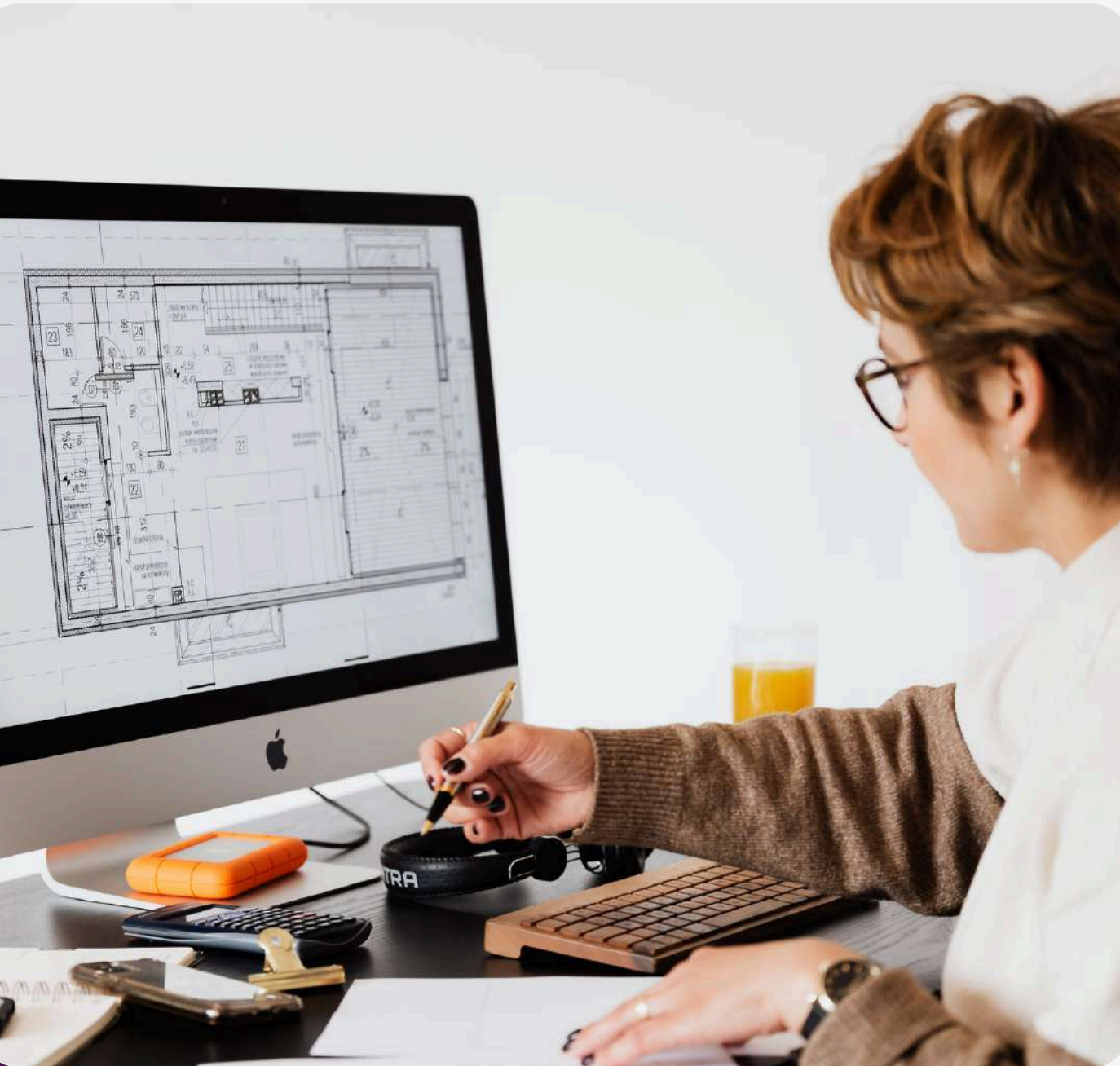


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# Data Analysis

**Hospital Sql Project**

**Name : Kanchana Gaira**

## SQL query

```
--1) Write an SQL query to find the total number of patients across all hospitals.  
SELECT Hospital_Name, SUM(Patients_Count) AS total_patient_count  
FROM patients  
GROUP BY Hospital_Name;
```

## Output

Data Output

Messages

Notifications

|    | hospital_name<br>character varying (100) | total_patient_count<br>bigint |
|----|------------------------------------------|-------------------------------|
| 1  | Metro_Hospital                           | 1121                          |
| 2  | Healing_Touch                            | 1059                          |
| 3  | City_Hospital                            | 843                           |
| 4  | Sunrise_Medical                          | 506                           |
| 5  | Apollo_Health                            | 1192                          |
| 6  | Global_Medicare                          | 986                           |
| 7  | Wellness_Clinic                          | 713                           |
| 8  | Fortis_Care                              | 895                           |
| 9  | Heritage_Hospital                        | 1179                          |
| 10 | Green_Valley_Hospital                    | 853                           |

## SQL query

```
--2) Retrieve the average count of doctors available in each hospital.  
SELECT Hospital_Name, AVG(Doctors_Count) AS avg_doctor_count  
FROM patients  
GROUP BY Hospital_Name;
```

## Output

Data Output

Messages

Notifications

|    | hospital_name<br>character varying (100) | avg_doctor_count<br>numeric |
|----|------------------------------------------|-----------------------------|
| 1  | Metro_Hospital                           | 27.4545454545454545         |
| 2  | Healing_Touch                            | 25.6428571428571429         |
| 3  | City_Hospital                            | 24.7000000000000000         |
| 4  | Sunrise_Medical                          | 28.6250000000000000         |
| 5  | Apollo_Health                            | 23.3636363636363636         |
| 6  | Global_Medicare                          | 28.7272727272727273         |
| 7  | Wellness_Clinic                          | 34.0000000000000000         |
| 8  | Fortis_Care                              | 24.5454545454545455         |
| 9  | Heritage_Hospital                        | 24.6666666666666667         |
| 10 | Green_Valley_Hospital                    | 28.8750000000000000         |

SQL query

```
--3) Find the top 3 hospital departments that have the highest number of patients
SELECT Department, SUM(Patients_Count)AS total_patients
FROM patients
GROUP BY Department
ORDER BY total_patients DESC LIMIT 3;
```

Output

| Data Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |                | Messages | Notifications |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------|----------|---------------|
| <div><div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div><div><div></div><div></div><div></div></div></div><div>SQL</div></div> |                        |                |          |               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | department             | total_patients |          |               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | character varying (50) | bigint         |          |               |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Urology                | 1766           |          |               |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Neurology              | 1229           |          |               |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ENT                    | 1064           |          |               |

## SQL query

```
--4) Identify the hospital that recorded the highest medical expenses
SELECT Hospital_Name, SUM (Medical_SExpenses) AS total_medicial_sexpenses
FROM patients
GROUP BY Hospital_Name
ORDER BY total_medicial_sexpenses DESC LIMIT 1;
```

## Output

| department<br>character varying (50) | total_patients<br>bigint |
|--------------------------------------|--------------------------|
| Urology                              | 1766                     |
| Neurology                            | 1229                     |
| ENT                                  | 1064                     |

## SQL query

```
--5) Calculate the average medical expenses per day for each hospital.
SELECT
    Hospital_Name,
    SUM(Medical_SExpenses) /
    SUM(Discharge_Date - Admission_Date) AS average_medical_expenses_per_day
FROM patients
GROUP BY Hospital_Name;
```

## Output

Data Output

Messages

Notifications

SQL

|    | hospital_name<br>character varying (100) | average_medical_expenses_per_day<br>numeric |
|----|------------------------------------------|---------------------------------------------|
| 1  | Metro_Hospital                           | 3245.2170103092783505                       |
| 2  | Healing_Touch                            | 3486.9114851485148515                       |
| 3  | City_Hospital                            | 4296.1535802469135802                       |
| 4  | Sunrise_Medical                          | 4315.7579629629629630                       |
| 5  | Apollo_Health                            | 3187.7596551724137931                       |
| 6  | Global_Medicare                          | 3507.8664893617021277                       |
| 7  | Wellness_Clinic                          | 2830.1261290322580645                       |
| 8  | Fortis_Care                              | 4385.4202597402597403                       |
| 9  | Heritage_Hospital                        | 2336.2378750000000000                       |
| 10 | Green_Valley_Hospital                    | 2615.9488709677419355                       |

SQL query

```
--6) Longest Hospital Stay
SELECT Hospital_Name, SUM(Discharge_Date - Admission_Date) AS total_stay_days
FROM patients
GROUP BY Hospital_Name ORDER BY total_stay_days LIMIT 1;
```

Output

| Data Output                                                                                                                                 |                                          |                           | Messages | Notifications |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------|----------|---------------|
| <div><div><div>≡+</div><div></div><div>▼</div><div></div><div>▼</div><div></div><div></div><div></div><div></div><div>SQL</div></div></div> |                                          |                           |          |               |
|                                                                                                                                             | hospital_name<br>character varying (100) | total_stay_days<br>bigint |          |               |
| 1                                                                                                                                           | Sunrise_Medical                          | 54                        |          |               |

## SQL query

```
--7 Total Patients Treated Per City
SELECT Location, SUM(Patients_Count ) AS total_patient
FROM patients
GROUP BY Location;
```

## Output

|    | location<br>character varying (50) | total_patient<br>bigint |
|----|------------------------------------|-------------------------|
| 1  | Bangalore                          | 955                     |
| 2  | Jaipur                             | 1505                    |
| 3  | Lucknow                            | 1264                    |
| 4  | Mumbai                             | 483                     |
| 5  | Kolkata                            | 596                     |
| 6  | Delhi                              | 421                     |
| 7  | Chennai                            | 840                     |
| 8  | Pune                               | 555                     |
| 9  | Hyderabad                          | 1261                    |
| 10 | Ahmedabad                          | 1467                    |

SQL query

```
-- 8) Calculate the average number of days patients spend in each department.  
SELECT Department, AVG(Discharge_Date - Admission_Date) AS average_days  
FROM patients  
GROUP BY Department;
```

Output

| Data Output                                                                                                       |                        |                    | Messages | Notifications |
|-------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|----------|---------------|
| <div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div>SQL</div></div> |                        |                    |          |               |
|                                                                                                                   | department             | average_days       |          |               |
|                                                                                                                   | character varying (50) | numeric            |          |               |
| 1                                                                                                                 | Dermatology            | 5.6000000000000000 |          |               |
| 2                                                                                                                 | Oncology               | 8.1111111111111111 |          |               |
| 3                                                                                                                 | ENT                    | 8.0833333333333333 |          |               |
| 4                                                                                                                 | Neurology              | 9.2500000000000000 |          |               |
| 5                                                                                                                 | Gynecology             | 7.6666666666666667 |          |               |
| 6                                                                                                                 | Pediatrics             | 9.1111111111111111 |          |               |
| 7                                                                                                                 | Orthopedics            | 7.1428571428571429 |          |               |
| 8                                                                                                                 | Cardiology             | 6.8571428571428571 |          |               |
| 9                                                                                                                 | Urology                | 8.7222222222222222 |          |               |
| 10                                                                                                                | General Medicine       | 7.4285714285714286 |          |               |

## SQL query

Process

Tools

Applications

--9) Find the department with the least number of patients.

```
SELECT Department, SUM(Patients_Count) AS total_patients
FROM patients
GROUP BY Department ORDER BY total_patients ASC LIMIT 1;
```

## Output

| Data Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                          | Messages                                                                                                       | Notifications |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------|
|          SQL |                                                                                                                          |                                                                                                                |               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | department<br>character varying (50)  | total_patients<br>bigint  |               |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cardiology                                                                                                               | 544                                                                                                            |               |

## SQL query

```
--10) Monthly Medical Expenses Report
SELECT DATE_TRUNC('MONTH', Admission_Date ) AS month,
SUM(Medical_SExpenses) AS total_Medical_SExpenses
FROM patients
GROUP BY month
ORDER BY month;
```

## Output

| Data Output                                                                         |                                                                                     | Messages                                                                            | Notifications                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
|  |  |  |  |
|  |  |  |  |
| month                                                                               | timestamp with time zone                                                            | total_medical_sexexpenses                                                           | numeric                                                                             |
| 1                                                                                   | 2023-01-01 00:00:00+05:30                                                           | 173971.54                                                                           |                                                                                     |
| 2                                                                                   | 2023-02-01 00:00:00+05:30                                                           | 301722.72                                                                           |                                                                                     |
| 3                                                                                   | 2023-03-01 00:00:00+05:30                                                           | 199247.42                                                                           |                                                                                     |
| 4                                                                                   | 2023-04-01 00:00:00+05:30                                                           | 88995.93                                                                            |                                                                                     |
| 5                                                                                   | 2023-05-01 00:00:00+05:30                                                           | 222986.72                                                                           |                                                                                     |
| 6                                                                                   | 2023-06-01 00:00:00+05:30                                                           | 165926.36                                                                           |                                                                                     |
| 7                                                                                   | 2023-07-01 00:00:00+05:30                                                           | 211527.13                                                                           |                                                                                     |
| 8                                                                                   | 2023-08-01 00:00:00+05:30                                                           | 181039.55                                                                           |                                                                                     |
| 9                                                                                   | 2023-09-01 00:00:00+05:30                                                           | 341284.23                                                                           |                                                                                     |
| 10                                                                                  | 2023-10-01 00:00:00+05:30                                                           | 158450.68                                                                           |                                                                                     |
| 11                                                                                  | 2023-11-01 00:00:00+05:30                                                           | 334370.33                                                                           |                                                                                     |
| 12                                                                                  | 2023-12-01 00:00:00+05:30                                                           | 337788.45                                                                           |                                                                                     |

# Thank You



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